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Open Geospatial Consortium

OGC INDOORGML 2.0 PART 2A – XML ENCODING

STANDARD

CANDIDATE SWG DRAFT

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I ABSTRACT

The IndoorGML 2.0 Part 2a – XML Encoding Standard presents the implementation-dependent Geography Markup Language (GML) XML schema encoding of the concepts defined by the IndoorGML 2.0 Part 1 – Conceptual Model Standard (OGC 22-045r5). The XML Encoding is compliant with GML version 3.2.1 as specified by OGC 07-036. The XML encoding of IndoorGML 2.0 defines a concrete implementation schema for representing indoor spatial information, including primal space, dual space, thematic layers, interlayer connections, and navigation-related indoor features. This encoding can be used to store and exchange 2D and/or 3D indoor space and indoor navigation network models in XML format as data sets or via web services.

This Part 2a of the Standard defines how the concepts developed in Part 1 are realized using XML Schema definitions. The XML encoding represents a full mapping of the Conceptual Model and is derived from the UML model following the UML-to-GML/XML encoding rules. Table 1 maps requirements classes from the IndoorGML 2.0 Conceptual Model into the implementation details documented in this standard.

Table 1 — Requirements classes mapping

CONCEPTUAL MODEL	SECTION	XML SCHEMA
IndoorGML Core	Clause 7	Clause 10.1
Navigation Extension	Clause 8	Clause 10.2

II KEYWORDS

The following are keywords to be used by search engines and document catalogues.

OGCDoc, OGC documents, indoor, navigation, IndoorGML, GML, XML

III PREFACE

In order to achieve consensus on the modeling indoor spaces and their neighborhood relationships to support indoor location-based services, a UML Conceptual Model, IndoorGML 2.0 Part 1, was approved as an OGC Standard (OGC 22-045r5) in June, 2025. This model covers geometric and semantic properties of indoor spaces relevant for indoor navigation. This IndoorGML 2.0 Part 2a: XML Encoding Standard defines how those concepts should be realized using XML schema format. The XML encoding is intended for software developers, data providers, standards implementers, and service providers who need a structured, standard, and machine-readable representation of IndoorGML datasets using XML/GML.

IV SECURITY CONSIDERATIONS

No security considerations have been made for this Standard.

V SUBMITTING ORGANIZATIONS

The following organizations submitted this Document to the Open Geospatial Consortium (OGC):

- The University of New South Wales
- Pusan National University
- CityGeometrix
- National Institute of Advanced Industrial Science and Technology
- SpatialAlgebra

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CONFORMANCE

2 CONFORMANCE

This standard defines implementation specification which specifies how the IndoorGML 2.0 Conceptual Model should be implemented using Extensible Markup Language (XML) and GML 3.2.1. The Standardization Target for this standard is:

- Implementations of the IndoorGML 2.0 Conceptual Model using XML encodings.

Conformance with this standard shall be checked using all the relevant tests specified in Annex A (normative) of this document. The framework, concepts, and methodology for testing, and the criteria to be achieved to claim conformance are specified in the OGC Compliance Testing Policies and Procedures and the OGC Compliance Testing web site.

In order to conform to this OGC® interface standard, a software implementation shall choose to implement:

- Any one of the conformance levels specified in Annex A (normative).

All requirements-classes and conformance-classes described in this document are owned by the standard(s) identified.

Table 2 — Conformance classes

CONFORMANCE CLASS	URI
IndoorGML Core	http://www.opengis.net/spec/indoorgml-2/2.0/xml/conf/core
Navigation Extension	http://www.opengis.net/spec/indoorgml-2/2.0/xml/conf/navi

1

SCOPE

1 SCOPE

The OGC® IndoorGML 2.0 Part 2a – XML Encoding Standard defines an XML encoding for IndoorGML 2.0 Part 1 — Conceptual Model. The encoding is implemented as an application schema of the Geography Markup Language version 3.2.1 (GML 3.2.1).

The IndoorGML XML Encoding Standard specifies:

- XML schema for IndoorGML 2.0 core module;
- XML schema for IndoorGML 2.0 navigation module;
- the geometry encoding rules based on GML 3.2.1 geometry objects; and
- a statement to which IndoorGML XML conformance classes an IndoorGML 2.0 instance conforms to.

IndoorGML XML encoding is an implementation encoding of IndoorGML 2.0 Part 1. Therefore, IndoorGML XML instances shall be interpreted according to the semantics defined in the IndoorGML 2.0 conceptual model.

3

NORMATIVE REFERENCES

3 NORMATIVE REFERENCES

The following documents are referred to in the text in such a way that some or all of their content constitutes requirements of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

Data elements and interchange formats – Information interchange – Representation of dates and times

Geographic information – Reference model – Part 1: Fundamentals

Geographic Information – Conceptual Schema Language

Geographic information – Conformance and testing

Geographic Information – Spatial Schema

Geographic Information – Rules for Application Schemas

Geographic information – Spatial referencing by coordinates

Geographic Information – Metadata – Part 1: Fundamentals

Geographic Information – Metadata – XML schema implementation

Carl Reed: OGC 04-084, *Topic 0 — Overview*. Open Geospatial Consortium (2005). https://docs.ogc.org/as/04-084/04-084/pdf

Cliff Kottman, Carl Reed: OGC 08-126, *Topic 5 — Features*. Open Geospatial Consortium (2009). https://docs.ogc.org/as/08-126/08-126/pdf

Cliff Kottman: OGC 99-108r2, *Topic 8 — Relationships Between Features*. Open Geospatial Consortium (1999). https://docs.ogc.org/as/99-108r2/99-108r2/pdf

Cliff Kottman: OGC 99-110, *Topic 10 — Feature Collections*. Open Geospatial Consortium (1999). https://docs.ogc.org/as/99-110/99-110/pdf

Clemens Portele: OGC 07-036, *OpenGIS Geography Markup Language (GML) Encoding Standard*. Open Geospatial Consortium (2007). https://docs.ogc.org/is/07-036/07-036/pdf

Sisi Zlatanova, Abdoulaye Diakite, Taehoon Kim, Ki-Joune Li : OGC 22-045r5, *OGC IndoorGML 2.0 Part 1 – Conceptual Model*. Open Geospatial Consortium (2025). http://www.opengis.net/doc/IS/indoorgml/2.0

4

TERMS AND DEFINITIONS

4 TERMS AND DEFINITIONS

This document uses the terms defined in OGC Policy Directive 49, which is based on the ISO/IEC Directives, Part 2, Rules for the structure and drafting of International Standards. In particular, the word “shall” (not “must”) is the verb form used to indicate a requirement to be strictly followed to conform to this document and OGC documents do not use the equivalent phrases in the ISO/IEC Directives, Part 2.

This document also uses terms defined in the OGC Standard for Modular specifications (OGC 08-131r3), also known as the ‘ModSpec’. The definitions of terms such as standard, specification, requirement, and conformance test are provided in the ModSpec.

For the purposes of this document, the following additional terms and definitions apply.

4.1. conceptual model

model that defines concepts of a universe of discourse

[**SOURCE:** ISO 19101-1]

4.2. conceptual schema

formal description of a conceptual model

[**SOURCE:** ISO 19101-1]

4.3. feature

abstraction of real world phenomena

Note 1 to entry: Features are objects that have an identity that allows for distinguishing features from each other. Features can have spatial and non-spatial properties that describe the features in more detail. Spatial properties are properties that have a geometry from ISO 19107 as data type.

Note 2 to entry: Features are denoted by the stereotype «FeatureType» in the IndoorGML 2.0 Conceptual Model.

[**SOURCE:** ISO 19101-1]

5

CONVENTIONS

5 CONVENTIONS

5.1. Identifiers

The normative provisions in this standard are denoted by the URI

http://www.opengis.net/spec/indoorgml-2/2.0/xml

All requirements and conformance tests that appear in this document are denoted by partial URIs which are relative to this base.

5.2. Use of XML terminology

The following terms and concepts are used as specified in the W3C XML specifications and OGC GML 3.2.1:

- “element” refers to an XML element tag;
- “attribute” refers to a name/value attribute on an XML element;
- “complexType” refers to an XML Schema complex type definition;
- “substitutionGroup” refers to the GML element substitution mechanism allowing specialized elements to substitute base GML feature types; and
- “property element” refers to an XML element that represents an association between features or objects.

5.3. UML Notation

The diagrams that appear in this standard are presented using the Unified Modeling Language (UML) static structure diagram.

In this standard, the following standard UML class stereotypes and data types are used:

- «Interface»: A definition of a set of operations supported by objects having this interface. An Interface class cannot contain any attributes.

- «DataType»: A descriptor of a set of values that lack identity.
- «CodeList»: A flexible enumeration that uses string values for expressing a list of potential values.

Standard data types used include CharacterString, Integer, Double, and Float.

6

OVERVIEW OF THE XML ENCODING

6 OVERVIEW OF THE XML ENCODING

The XML encoding of IndoorGML 2.0 implements the conceptual classes of IndoorGML 2.0 Part 1 as GML-compliant XML elements and types. The conceptual model remains normative for semantics, while this document defines the XML schema representation.

6.1. Design Principle

The XML encoding is based on the following encoding principles.

6.1.1. Conceptual model preservation

The XML encoding of the IndoorGML core module preserves the IndoorGML 2.0 core conceptual model structure, as shown in Figure 1.

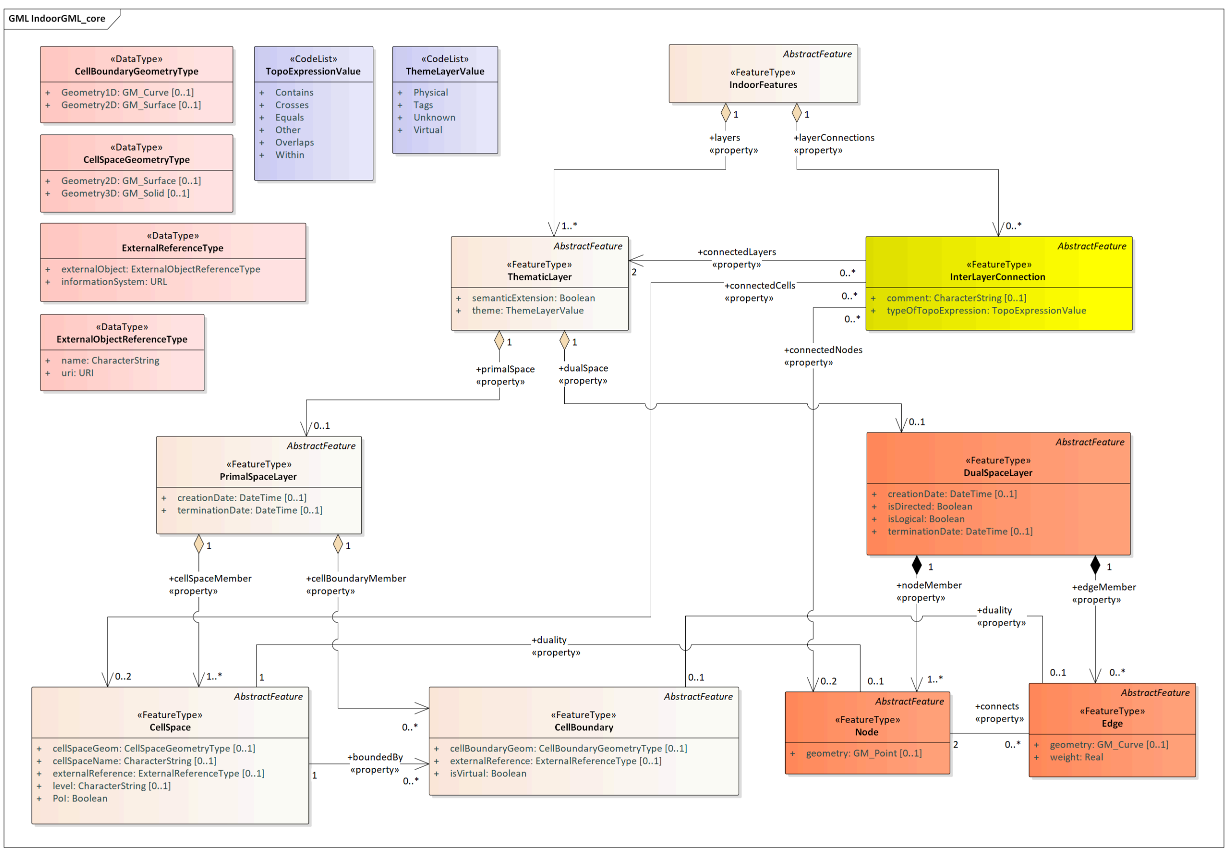

Figure 1 — IndoorGML 2.0 core module structure

Each feature class is mapped to an XML element inheriting from or substituting GML feature elements (e.g. gml:AbstractFeature). For example, the <CellSpace> element substitutes gml:AbstractFeature and extends gml:AbstractFeatureType to represent cell spaces.

The XML encoding of the IndoorGML navigation module preserves the IndoorGML 2.0 navigation conceptual model structure, as shown in Figure 2.

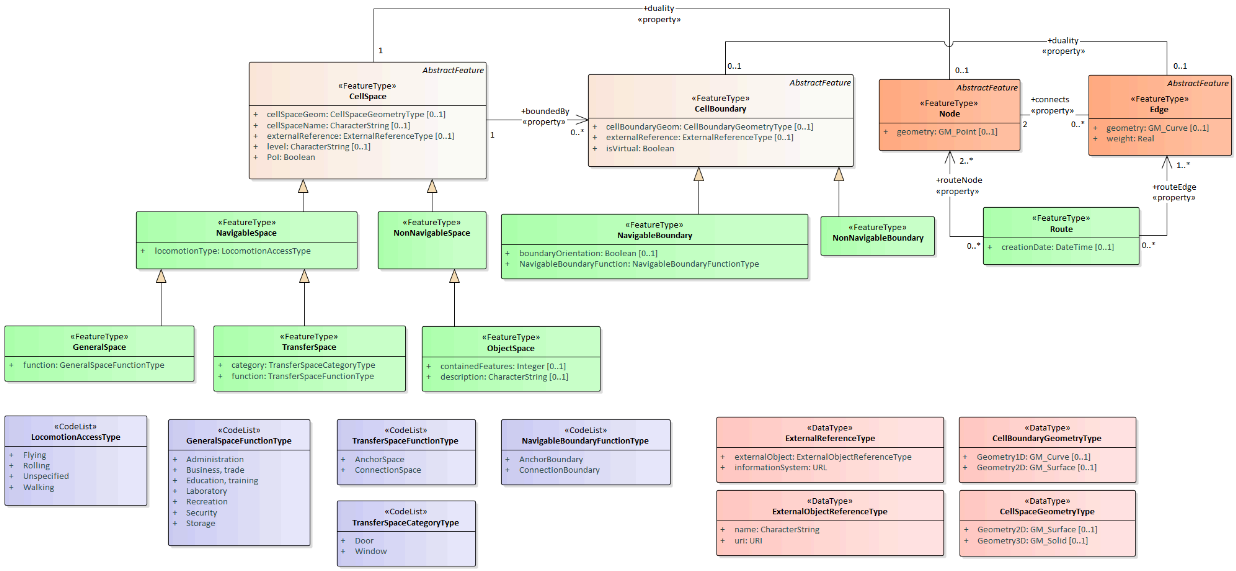

Figure 2 — IndoorGML 2.0 navigation module structure

Navigation feature classes extend core elements using GML inheritance patterns, implementing substitutions where appropriate. For example, <NavigableSpace> substitutes <CellSpace> and extends its complex type definition.

6.1.2. XML Schema design workflow

The XML encoding schema is derived from the IndoorGML 2.0 UML model and is designed on the basis of OGC GML 3.2.1. XML schema (.xsd) documents are generated from the UML model, establishing elements and complex types for the Core and Navigation modules. Typical patterns include converting associations into GML property types using xlink:href references (for example, the duality link from CellSpace to Node), mapping attributes to native XML schema elements (such as creationDate of type xs:dateTime), and representing UML code lists as XML simple types restricting xs:string with enumeration constraints.

7

INDOORXML CORE

7 INDOORXML CORE

REQUIREMENTS CLASS 1	
Target type	Implementation Specification
Dependency	http://www.opengis.net/spec/indoorgml/2.0/req
Dependency	http://www.opengis.net/spec/gml/3.2.1/req
REQUIREMENT 1-1	
REQUIREMENT 1-2	
REQUIREMENT 1-3	
REQUIREMENT 1-4	
REQUIREMENT 1-5	
REQUIREMENT 1-6	
REQUIREMENT 1-7	
REQUIREMENT 1-8	
REQUIREMENT 1-9	

7.1. General requirements for encoding standard

REQUIREMENT 1: REQUIREMENT FOR GLOBAL XML SCHEMA VALIDATION	
A	An instance XML document representing an IndoorGML 2.0 dataset SHALL be well-formed XML and validate against the schemas defined in this standard.

7.2. CellSpace

REQUIREMENT 2: REQUIREMENT FOR CELLSPACE XML ELEMENT	
A	An instance CellSpace XML element SHALL conform to the schema structure defined in the IndoorGML 2.0 Core XML Schema (IndoorGML_core.xsd).

```
<element name="CellSpace" substitutionGroup="gml:AbstractFeature" type="core:CellSpaceType"/>
 <complexType name="CellSpaceType">
  <complexContent>
   <extension base="gml:AbstractFeatureType">
    <sequence>
     <element minOccurs="0" name="cellSpaceGeom" type="core:CellSpaceGeometryPropertyType"/>
     <element minOccurs="0" name="externalReference" type="core:ExternalReferencePropertyType"/>
     <element minOccurs="0" name="level" type="string"/>
     <element name="PoI" type="boolean"/>
     <element minOccurs="0" name="cellSpaceName" type="string"/>
     <element minOccurs="0" name="duality" type="core:NodePropertyType"/>
     <element maxOccurs="unbounded" minOccurs="0" name="boundedBy" type="core:
CellBoundaryPropertyType"/>
    </sequence>
   </extension>
  </complexContent>
 </complexType>
 <complexType name="CellSpacePropertyType">
  <sequence minOccurs="0">
   <element ref="core:CellSpace"/>
  </sequence>
  <attributeGroup ref="gml:AssociationAttributeGroup"/>
  <attributeGroup ref="gml:OwnershipAttributeGroup"/>
 </complexType>
 <element name="CellSpaceGeometry" substitutionGroup="gml:AbstractObject" type="core:
CellSpaceGeometryType"/>
 <complexType name="CellSpaceGeometryType">
  <sequence>
   <element minOccurs="0" name="Geometry3D" type="gml:SolidPropertyType"/>
   <element minOccurs="0" name="Geometry2D" type="gml:SurfacePropertyType"/>
  </sequence>
 </complexType>
 <complexType name="CellSpaceGeometryPropertyType">
  <sequence>
   <element ref="core:CellSpaceGeometry"/>
  </sequence>
 </complexType>
```

```
<core:CellSpace gml:id="CS_1">
  <!-- mandatory property -->
  <core:PoI>false</core:PoI>

  <!-- optional properties -->
  <core:cellSpaceGeom>
    <core:Geometry3D>
      <gml:Solid gml:id="CSG_CS_1">
        ...
      </gml:Solid>
    </core:Geometry3D>
  </core:cellSpaceGeom>
```

```xml
    <core:duality xlink:href="#ND_1"/>
    <core:level>3rd Floor</core:level>
    <core:cellSpaceName>Locker Room Lv3</core:cellSpaceName>
</core:CellSpace>
```

7.3. CellBoundary

REQUIREMENT 3: REQUIREMENT FOR CELLBOUNDARY XML ELEMENT	
A	An instance CellBoundary XML element SHALL conform to the schema structure defined in the IndoorGML 2.0 Core XML Schema (IndoorGML_core.xsd).

```xml
<element name="CellBoundary" substitutionGroup="gml:AbstractFeature" type="core:CellBoundaryType"/>
 <complexType name="CellBoundaryType">
  <complexContent>
   <extension base="gml:AbstractFeatureType">
    <sequence>
     <element minOccurs="0" name="cellBoundaryGeom" type="core:CellBoundaryGeometryPropertyType"/>
     <element minOccurs="0" name="externalReference" type="core:ExternalReferencePropertyType"/>
     <element name="isVirtual" type="boolean"/>
     <element minOccurs="0" name="duality" type="core:EdgePropertyType"/>
    </sequence>
   </extension>
  </complexContent>
 </complexType>
 <complexType name="CellBoundaryPropertyType">
  <sequence minOccurs="0">
   <element ref="core:CellBoundary"/>
  </sequence>
  <attributeGroup ref="gml:AssociationAttributeGroup"/>
  <attributeGroup ref="gml:OwnershipAttributeGroup"/>
 </complexType>
 <element name="CellBoundaryGeometry" substitutionGroup="gml:AbstractObject" type="core:
CellBoundaryGeometryType"/>
 <complexType name="CellBoundaryGeometryType">
  <sequence>
   <element minOccurs="0" name="Geometry2D" type="gml:SurfacePropertyType"/>
   <element minOccurs="0" name="Geometry1D" type="gml:CurvePropertyType"/>
  </sequence>
 </complexType>
 <complexType name="CellBoundaryGeometryPropertyType">
  <sequence>
   <element ref="core:CellBoundaryGeometry"/>
  </sequence>
 </complexType>
```

```xml
<core:CellBoundary gml:id="CB_1">
  <!-- mandatory property -->
  <core:isVirtual>false</core:isVirtual>

  <!-- optional properties -->
  <core:cellBoundaryGeom>
    <core:Geometry2D>
      <gml:Surface gml:id="CBG_CB_1">
        ...
      </gml:Surface>
```

```
    </core:Geometry2D>
  </core:cellBoundaryGeom>
  <core:duality xlink:href="#EDG_1"/>
</core:CellBoundary>
```

7.4. PrimalSpaceLayer

REQUIREMENT 4: REQUIREMENT FOR PRIMALSPACELAYER XML ELEMENT	
A	An instance PrimalSpaceLayer XML element SHALL conform to the schema structure defined in the IndoorGML 2.0 Core XML Schema (IndoorGML_core.xsd).

```
<element name="PrimalSpaceLayer" substitutionGroup="gml:AbstractFeature" type="core:PrimalSpaceLayerType"/
>
 <complexType name="PrimalSpaceLayerType">
  <complexContent>
   <extension base="gml:AbstractFeatureType">
    <sequence>
     <element minOccurs="0" name="creationDate" type="dateTime"/>
     <element minOccurs="0" name="terminationDate" type="dateTime"/>
     <element maxOccurs="unbounded" name="cellSpaceMember">
      <complexType>
       <complexContent>
        <extension base="gml:AbstractFeatureMemberType">
         <sequence minOccurs="0">
          <element ref="core:CellSpace"/>
         </sequence>
         <attributeGroup ref="gml:AssociationAttributeGroup"/>
        </extension>
       </complexContent>
      </complexType>
     </element>
     <element maxOccurs="unbounded" minOccurs="0" name="cellBoundaryMember">
      <complexType>
       <complexContent>
        <extension base="gml:AbstractFeatureMemberType">
         <sequence minOccurs="0">
          <element ref="core:CellBoundary"/>
         </sequence>
         <attributeGroup ref="gml:AssociationAttributeGroup"/>
        </extension>
       </complexContent>
      </complexType>
     </element>
    </sequence>
   </extension>
  </complexContent>
 </complexType>
```

```
<core:PrimalSpaceLayer gml:id="PSL_1">
  <!-- mandatory properties (at least one CellSpace) -->
  <core:cellSpaceMember>
    <core:CellSpace gml:id="CS_1">...</core:CellSpace>
  </core:cellSpaceMember>
  <!-- optional properties -->
  <core:cellBoundaryMember>
```

```
    <core:CellBoundary gml:id="CB_1">...</core:CellBoundary>
  </core:cellBoundaryMember>
  <core:creationDate>2025-08-06T18:39:00Z</core:creationDate>
</core:PrimalSpaceLayer>
```

7.5. Node

REQUIREMENT 5: REQUIREMENT FOR NODE XML ELEMENT	
A	An instance Node XML element SHALL conform to the schema structure defined in the IndoorGML 2.0 Core XML Schema (IndoorGML_core.xsd).

```
<element name="Node" substitutionGroup="gml:AbstractFeature" type="core:NodeType"/>
 <complexType name="NodeType">
  <complexContent>
   <extension base="gml:AbstractFeatureType">
    <sequence>
     <element minOccurs="0" name="geometry" type="gml:PointPropertyType"/>
     <element name="duality" type="core:CellSpacePropertyType"/>
     <element maxOccurs="unbounded" minOccurs="0" name="connects" type="core:EdgePropertyType"/>
    </sequence>
   </extension>
  </complexContent>
 </complexType>
```

```
<core:Node gml:id="ND_1">
  <!-- mandatory property -->
  <core:duality xlink:href="#CS_1"/>
  <!-- optional properties -->
  <core:geometry>
    <gml:Point gml:id="G_ND_1">
      <gml:pos>445536.499779417 5444906.24858758 -2.02</gml:pos>
    </gml:Point>
  </core:geometry>
  <core:connects xlink:href="#EDG_1"/>
</core:Node>
```

7.6. Edge

REQUIREMENT 6: REQUIREMENT FOR EDGE XML ELEMENT	
A	An instance Edge XML element SHALL conform to the schema structure defined in the IndoorGML 2.0 Core XML Schema (IndoorGML_core.xsd).

```
<element name="Edge" substitutionGroup="gml:AbstractFeature" type="core:EdgeType"/>
 <complexType name="EdgeType">
  <complexContent>
   <extension base="gml:AbstractFeatureType">
```

```
    <sequence>
     <element minOccurs="0" name="geometry" type="gml:CurvePropertyType"/>
     <element name="weight" type="double"/>
     <element minOccurs="0" name="duality" type="core:CellBoundaryPropertyType"/>
     <element maxOccurs="2" minOccurs="2" name="connects" type="core:NodePropertyType"/>
    </sequence>
   </extension>
  </complexContent>
 </complexType>
```

```
<core:Edge gml:id="EDG_1">
  <!-- mandatory properties -->
  <core:connects xlink:href="#ND_1"/>
  <core:connects xlink:href="#ND_2"/>
  <!-- optional properties -->
  <core:geometry>
    <gml:LineString gml:id="G_EDG_1">
      <gml:pos>445536.499779417 5444906.24858758 -2.02</gml:pos>
      <gml:pos>445537.914644321 5444904.59001251 -2.02</gml:pos>
    </gml:LineString>
  </core:geometry>
  <core:duality xlink:href="#CB_1"/>
</core:Edge>
```

7.7. DualSpaceLayer

REQUIREMENT 7: REQUIREMENT FOR DUALSPACELAYER XML ELEMENT	
A	An instance DualSpaceLayer XML element SHALL conform to the schema structure defined in the IndoorGML 2.0 Core XML Schema (IndoorGML_core.xsd).

```
<element name="DualSpaceLayer" substitutionGroup="gml:AbstractFeature" type="core:DualSpaceLayerType"/>
 <complexType name="DualSpaceLayerType">
  <complexContent>
   <extension base="gml:AbstractFeatureType">
    <sequence>
     <element name="isLogical" type="boolean"/>
     <element minOccurs="0" name="creationDate" type="dateTime"/>
     <element minOccurs="0" name="terminationDate" type="dateTime"/>
     <element name="isDirected" type="boolean"/>
     <element maxOccurs="unbounded" minOccurs="0" name="edgeMember">
      <complexType>
       <complexContent>
        <extension base="gml:AbstractFeatureMemberType">
         <sequence minOccurs="0">
          <element ref="core:Edge"/>
         </sequence>
         <attributeGroup ref="gml:AssociationAttributeGroup"/>
        </extension>
       </complexContent>
      </complexType>
     </element>
     <element maxOccurs="unbounded" name="nodeMember">
      <complexType>
       <complexContent>
        <extension base="gml:AbstractFeatureMemberType">
```

```xml
          <sequence minOccurs="0">
           <element ref="core:Node"/>
          </sequence>
          <attributeGroup ref="gml:AssociationAttributeGroup"/>
         </extension>
        </complexContent>
       </complexType>
      </element>
     </sequence>
    </extension>
   </complexContent>
  </complexType>
```

```xml
<core:DualSpaceLayer gml:id="DSL_1">
  <!-- mandatory properties -->
  <core:isLogical>true</core:isLogical>
  <core:isDirected>false</core:isDirected>
  <core:nodeMember>
    <core:Node gml:id="ND_1">...</core:Node>
  </core:nodeMember>
  <!-- optional properties -->
  <core:edgeMember>
    <core:Edge gml:id="EDG_1">...</core:Edge>
  </core:edgeMember>
  <core:creationDate>2025-08-06T18:39:00Z</core:creationDate>
</core:DualSpaceLayer>
```

7.8. InterLayerConnection

REQUIREMENT 8: REQUIREMENT FOR INTERLAYERCONNECTION XML ELEMENT	
A	An instance InterLayerConnection XML element SHALL conform to the schema structure defined in the IndoorGML 2.0 Core XML Schema (IndoorGML_core.xsd).

```xml
<element name="InterLayerConnection" substitutionGroup="gml:AbstractFeature" type="core:
InterLayerConnectionType"/>
 <complexType name="InterLayerConnectionType">
  <complexContent>
   <extension base="gml:AbstractFeatureType">
    <sequence>
     <element name="typeOfTopoExpression" type="core:TopoExpressionValueType"/>
     <element minOccurs="0" name="comment" type="string"/>
     <element maxOccurs="2" minOccurs="2" name="connectedLayers" type="core:ThematicLayerPropertyType"/>
     <element maxOccurs="2" minOccurs="0" name="connectedCells" type="core:CellSpacePropertyType"/>
     <element maxOccurs="2" minOccurs="0" name="connectedNodes" type="core:NodePropertyType"/>
    </sequence>
   </extension>
  </complexContent>
 </complexType>
```

```xml
<core:InterLayerConnection gml:id="ILC_1">
  <!-- mandatory properties -->
  <core:typeOfTopoExpression>Contains</core:typeOfTopoExpression>
  <core:connectedLayers>
    <core:ThematicLayer gml:id="TL_1">...</core:ThematicLayer>
    <core:ThematicLayer gml:id="TL_2">...</core:ThematicLayer>
```

```
  </core:connectedLayers>
  <!-- optional properties -->
  <core:comment>TL_1 describes the room spaces and contains TL_2</core:comment>
  <core:connectedCells xlink:href="#PS1_CS_1"/>
  <core:connectedCells xlink:href="#PS2_CS_1"/>
</core:InterLayerConnection>
```

7.9. ThematicLayer

REQUIREMENT 9: REQUIREMENT FOR THEMATICLAYER XML ELEMENT	
A	An instance ThematicLayer XML element SHALL conform to the schema structure defined in the IndoorGML 2.0 Core XML Schema (IndoorGML_core.xsd).

```
<element name="ThematicLayer" substitutionGroup="gml:AbstractFeature" type="core:ThematicLayerType"/>
 <complexType name="ThematicLayerType">
  <complexContent>
   <extension base="gml:AbstractFeatureType">
    <sequence>
     <element name="semanticExtension" type="boolean"/>
     <element name="theme" type="core:ThemeLayerValueType"/>
     <element minOccurs="0" name="primalSpace">
      <complexType>
       <complexContent>
        <extension base="gml:AbstractFeatureMemberType">
         <sequence minOccurs="0">
          <element ref="core:PrimalSpaceLayer"/>
         </sequence>
         <attributeGroup ref="gml:AssociationAttributeGroup"/>
        </extension>
       </complexContent>
      </complexType>
     </element>
     <element minOccurs="0" name="dualSpace">
      <complexType>
       <complexContent>
        <extension base="gml:AbstractFeatureMemberType">
         <sequence minOccurs="0">
          <element ref="core:DualSpaceLayer"/>
         </sequence>
         <attributeGroup ref="gml:AssociationAttributeGroup"/>
        </extension>
       </complexContent>
      </complexType>
     </element>
    </sequence>
   </extension>
  </complexContent>
 </complexType>
```

```
<core:ThematicLayer gml:id="TL_1">
  <!-- mandatory properties -->
  <core:semanticExtension>true</core:semanticExtension>
  <core:theme>Physical</core:theme>
  <!-- optional properties -->
  <core:primalSpace xlink:href="#PSL_1"/>
```

```
  <core:dualSpace xlink:href="#DSL_1"/>
</core:ThematicLayer>
```

8

INDOORXML NAVIGATION

8 INDOORXML NAVIGATION

REQUIREMENTS CLASS 2	
Target type	Implementation Specification
Dependency	http://www.opengis.net/spec/indoorgml/2.0/req
Dependency	http://www.opengis.net/spec/indoorgml-2/2.0/xml/req/core
REQUIREMENT 2-1	
REQUIREMENT 2-2	
REQUIREMENT 2-3	
REQUIREMENT 2-4	
REQUIREMENT 2-5	
REQUIREMENT 2-6	
REQUIREMENT 2-7	
REQUIREMENT 2-8	
REQUIREMENT 2-9	

8.1. General requirements for encoding standards

REQUIREMENT 10: REQUIREMENT FOR NAVIGATION MODULE XML SCHEMA VALIDATION	
A	An instance XML document representing an IndoorGML 2.0 Navigation module dataset SHALL conform to the schemas defined in this standard.

8.2. NavigableSpace

REQUIREMENT 11: REQUIREMENT FOR NAVIGABLESPACE XML ELEMENT	
A	An instance NavigableSpace XML element SHALL conform to the schema structure defined in the IndoorGML 2.0 Navigation XML Schema (IndoorGML_navi.xsd).

```
<element name="NavigableSpace" substitutionGroup="core:CellSpace" type="navi:NavigableSpaceType"/>
 <complexType name="NavigableSpaceType">
  <complexContent>
   <extension base="core:CellSpaceType">
    <sequence>
     <element name="locomotionType" type="navi:LocomotionAccessType"/>
    </sequence>
   </extension>
  </complexContent>
 </complexType>
```

```
<navi:NavigableSpace gml:id="NS_1">
  <!-- mandatory property -->
  <navi:locomotionType>Walking</navi:locomotionType>
</navi:NavigableSpace>
```

8.3. GeneralSpace

REQUIREMENT 12: REQUIREMENT FOR GENERALSPACE XML ELEMENT	
A	An instance GeneralSpace XML element SHALL conform to the schema structure defined in the IndoorGML 2.0 Navigation XML Schema (IndoorGML_navi.xsd).

```
<element name="GeneralSpace" substitutionGroup="navi:NavigableSpace" type="navi:GeneralSpaceType"/>
 <complexType name="GeneralSpaceType">
  <complexContent>
   <extension base="navi:NavigableSpaceType">
    <sequence>
     <element name="function" type="navi:GeneralSpaceFunctionType"/>
    </sequence>
   </extension>
  </complexContent>
 </complexType>
```

```
<navi:GeneralSpace gml:id="GS_1">
  <!-- mandatory property -->
  <navi:function>Storage</navi:function>
</navi:GeneralSpace>
```

8.4. TransferSpace

REQUIREMENT 13: REQUIREMENT FOR TRANSFERSPACE XML ELEMENT	
A	An instance TransferSpace XML element SHALL conform to the schema structure defined in the IndoorGML 2.0 Navigation XML Schema (IndoorGML_navi.xsd).

```
<element name="TransferSpace" substitutionGroup="navi:NavigableSpace" type="navi:TransferSpaceType"/>
<complexType name="TransferSpaceType">
 <complexContent>
  <extension base="navi:NavigableSpaceType">
   <sequence>
    <element name="function" type="navi:TransferSpaceFunctionType"/>
    <element name="type" type="navi:TransferSpaceType"/>
   </sequence>
  </extension>
 </complexContent>
</complexType>
```

```
<navi:TransferSpace gml:id="TS_1">
  <!-- mandatory properties -->
  <navi:category>Door</navi:category>
  <navi:function>ConnectionSpace</navi:function>
</navi:TransferSpace>
```

8.5. NavigableBoundary

REQUIREMENT 14: REQUIREMENT FOR NAVIGABLEBOUNDARY XML ELEMENT	
A	An instance NavigableBoundary XML element SHALL conform to the schema structure defined in the IndoorGML 2.0 Navigation XML Schema (IndoorGML_navi.xsd).

```
<element name="NavigableBoundary" substitutionGroup="core:CellBoundary" type="navi:
NavigableBoundaryType"/>
<complexType name="NavigableBoundaryType">
 <complexContent>
  <extension base="core:CellBoundaryType">
   <sequence>
    <element name="NavigableBoundaryFunction" type="navi:NavigableBoundaryFunctionType"/>
    <element minOccurs="0" name="boundaryOrientation" type="boolean"/>
   </sequence>
  </extension>
 </complexContent>
</complexType>
```

```
<navi:NavigableBoundary gml:id="NB_1">
  <!-- mandatory property -->
  <navi:NavigableBoundaryFunction>AnchorBoundary</navi:NavigableBoundaryFunction>
  <!-- Optional property -->
  <navi:boundaryOrientation>false</navi:boundaryOrientation>
```

```
</navi:NavigableBoundary>
```

8.6. NonNavigableSpace

REQUIREMENT 15: REQUIREMENT FOR NONNAVIGABLESPACE XML ELEMENT	
A	An instance NonNavigableSpace XML element SHALL conform to the schema structure defined in the IndoorGML 2.0 Navigation XML Schema (IndoorGML_navi.xsd).

```
<element name="NonNavigableSpace" substitutionGroup="core:CellSpace" type="navi:NonNavigableSpaceType"/>
<complexType name="NonNavigableSpaceType">
 <complexContent>
  <extension base="core:CellSpaceType">
   <sequence/>
  </extension>
 </complexContent>
</complexType>
```

```
<navi:NonNavigableSpace gml:id="NNS_1">
  <!-- no property -->
</navi:NonNavigableSpace>
```

8.7. ObjectSpace

REQUIREMENT 16: REQUIREMENT FOR OBJECTSPACE XML ELEMENT	
A	An instance ObjectSpace XML element SHALL conform to the schema structure defined in the IndoorGML 2.0 Navigation XML Schema (IndoorGML_navi.xsd).

```
<element name="ObjectSpace" substitutionGroup="navi:NonNavigableSpace" type="navi:ObjectSpaceType"/>
<complexType name="ObjectSpaceType">
 <complexContent>
  <extension base="navi:NonNavigableSpaceType">
   <sequence>
    <element minOccurs="0" name="description" type="string"/>
    <element minOccurs="0" name="containedFeatures" type="integer"/>
   </sequence>
  </extension>
 </complexContent>
</complexType>
```

```
<navi:ObjectSpace gml:id="OS_1">
  <!-- Optional properties -->
  <navi:containedFeatures>3</navi:containedFeatures>
  <navi:description>Living room furniture</navi:description>
</navi:ObjectSpace>
```

8.8. NonNavigableBoundary

REQUIREMENT 17: REQUIREMENT FOR NONNAVIGABLEBOUNDARY XML ELEMENT	
A	An instance NonNavigableBoundary XML element SHALL conform to the schema structure defined in the IndoorGML 2.0 Navigation XML Schema (IndoorGML_navi.xsd).

```
<element name="NonNavigableBoundary" substitutionGroup="core:CellBoundary" type="navi:
NonNavigableBoundaryType"/>
<complexType name="NonNavigableBoundaryType">
 <complexContent>
  <extension base="core:CellBoundaryType">
   <sequence/>
  </extension>
 </complexContent>
</complexType>
```

```
<navi:NonNavigableBoundary gml:id="NNB_1">
  <!-- no property -->
</navi:NonNavigableBoundary>
```

8.9. Route

REQUIREMENT 18: REQUIREMENT FOR ROUTE XML ELEMENT	
A	An instance Route XML element SHALL conform to the schema structure defined in the IndoorGML 2.0 Navigation XML Schema (IndoorGML_navi.xsd).

```
<element name="Route" substitutionGroup="gml:AbstractFeature" type="navi:RouteType"/>
<complexType name="RouteType">
 <complexContent>
  <extension base="gml:AbstractFeatureType">
   <sequence>
    <element minOccurs="0" name="creationDate" type="dateTime"/>
    <element maxOccurs="unbounded" minOccurs="2" name="routeNode" type="core:NodePropertyType"/>
    <element maxOccurs="unbounded" name="routeEdge" type="core:EdgePropertyType"/>
   </sequence>
  </extension>
 </complexContent>
</complexType>
```

```
<navi:Route gml:id="RT_1">
  <!-- mandatory properties (at least 2 nodes and 1 edge) -->
  <navi:routeNode xlink:href="#ND_1"/>
  <navi:routeNode xlink:href="#ND_2"/>
  <navi:routeEdge xlink:href="#EDG_1"/>
  <!-- Optional property -->
  <navi:creationDate>2025-10-20T16:44:00Z</navi:creationDate>
</navi:Route>
```

9

ABSTRACT TEST SUITES FOR INDOORGML INSTANCE DOCUMENTS

9 ABSTRACT TEST SUITES FOR INDOORGML INSTANCE DOCUMENTS

9.1. Test Cases for mandatory conformance requirements

9.1.1. Valid IndoorGML instance document

- **Test purpose:** Verify the validity of the IndoorGML instance document against the XML Schema definition of each IndoorGML module that is part of the IndoorGML profile employed by the instance document. This may be any combination of IndoorGML extension modules in conjunction with the IndoorGML core module.
- **Test method:** Validate the IndoorGML XML instance document against the XML Schema definitions of all employed IndoorGML modules. The process may be using an appropriate software tool for validation or be a manual process that checks all relevant definitions from the respective XML Schema specification of the employed IndoorGML modules.
- **Reference:** Annex B
- **Test type:** Basic Test

10 XML SCHEMA FOR INDOORGML 2

10 XML SCHEMA FOR INDOORGML 2

In addition to this document, this standard includes some normative XML Schema Documents. These XML Schema Documents are included in this document. These XML Schema Documents are also posted online at the URL http://schemas.opengis.net/indoorgml/2.0. In the event of a discrepancy between this document and online versions of the XML Schema Document files (posted at schemas.opengis.net), the online schema files SHALL be considered authoritative.

10.1. IndoorGML Core Module

```
<?xml version="1.0" encoding="UTF-8"?><schema xmlns="http://www.w3.org/2001/XMLSchema" xmlns:IndoorGML=
"http://www.opengis.net/indoorgml/2.0/navi" xmlns:gmd="http://www.isotc211.org/2005/gmd" xmlns:gml=
"http://www.opengis.net/gml/3.2" elementFormDefault="qualified" targetNamespace="http://www.opengis.net/
indoorgml/2.0/core" version="2.0">
 <annotation>
  <appinfo>
   <gmlProfileSchema xmlns="http://www.opengis.net/gml/3.2">IndoorGML</gmlProfileSchema>
  </appinfo>
 </annotation>
 <import namespace="http://www.isotc211.org/2005/gmd" schemaLocation="http://schemas.opengis.net/iso/19139
/20070417/gmd/gmd.xsd"/>
 <import namespace="http://www.opengis.net/gml/3.2" schemaLocation="http://schemas.opengis.net/gml/3.2.1/
gml.xsd"/>
 <import namespace="http://www.opengis.net/indoorgml/2.0/navi"/>
 <!--XML Schema document created by ShapeChange - http://shapechange.net/-->
 <element name="AbstractFeature" substitutionGroup="gml:AbstractFeature" type="IndoorGML:
AbstractFeatureType"/>
 <complexType name="AbstractFeatureType">
  <complexContent>
   <extension base="gml:AbstractFeatureType">
    <sequence/>
   </extension>
  </complexContent>
 </complexType>
 <complexType name="AbstractFeaturePropertyType">
  <sequence minOccurs="0">
   <element ref="IndoorGML:AbstractFeature"/>
  </sequence>
  <attributeGroup ref="gml:AssociationAttributeGroup"/>
  <attributeGroup ref="gml:OwnershipAttributeGroup"/>
 </complexType>
 <element name="CellBoundary" substitutionGroup="gml:AbstractFeature" type="IndoorGML:CellBoundaryType"/>
 <complexType name="CellBoundaryType">
  <complexContent>
   <extension base="gml:AbstractFeatureType">
    <sequence>
     <element minOccurs="0" name="cellBoundaryGeom" type="IndoorGML:CellBoundaryGeometryPropertyType"/
>
     <element minOccurs="0" name="externalReference" type="IndoorGML:ExternalReferencePropertyType"/>
     <element name="isVirtual" type="boolean"/>
     <element minOccurs="0" name="duality" type="IndoorGML:EdgePropertyType">
      <annotation>
       <documentation>Corresponding cell boundary in the primal space</documentation>
```

```
          <appinfo>
            <reversePropertyName xmlns="http://www.opengis.net/gml/3.2">IndoorGML:
duality</reversePropertyName>
          </appinfo>
        </annotation>
      </element>
    </sequence>
  </extension>
 </complexContent>
</complexType>
<complexType name="CellBoundaryPropertyType">
  <sequence minOccurs="0">
    <element ref="IndoorGML:CellBoundary"/>
  </sequence>
  <attributeGroup ref="gml:AssociationAttributeGroup"/>
  <attributeGroup ref="gml:OwnershipAttributeGroup"/>
</complexType>
<element name="CellBoundaryGeometry" substitutionGroup="gml:AbstractObject" type="IndoorGML:
CellBoundaryGeometryType"/>
<complexType name="CellBoundaryGeometryType">
  <sequence>
    <element minOccurs="0" name="Geometry2D" type="gml:SurfacePropertyType"/>
    <element minOccurs="0" name="Geometry1D" type="gml:CurvePropertyType"/>
  </sequence>
</complexType>
<complexType name="CellBoundaryGeometryPropertyType">
  <sequence>
    <element ref="IndoorGML:CellBoundaryGeometry"/>
  </sequence>
</complexType>
<element name="CellSpace" substitutionGroup="gml:AbstractFeature" type="IndoorGML:CellSpaceType"/>
<complexType name="CellSpaceType">
  <complexContent>
    <extension base="gml:AbstractFeatureType">
      <sequence>
        <element minOccurs="0" name="cellSpaceGeom" type="IndoorGML:CellSpaceGeometryPropertyType"/>
        <element minOccurs="0" name="externalReference" type="IndoorGML:ExternalReferencePropertyType"/>
        <element minOccurs="0" name="level" type="string"/>
        <element name="PoI" type="boolean"/>
        <element minOccurs="0" name="cellSpaceName" type="string"/>
        <element minOccurs="0" name="duality" type="IndoorGML:NodePropertyType">
          <annotation>
            <documentation>Corresponding cell space in the primal space</documentation>
            <appinfo>
              <reversePropertyName xmlns="http://www.opengis.net/gml/3.2">IndoorGML:
duality</reversePropertyName>
            </appinfo>
          </annotation>
        </element>
        <element maxOccurs="unbounded" minOccurs="0" name="boundedBy" type="IndoorGML:
CellBoundaryPropertyType">
          <annotation>
            <documentation>Cell boundaries that are bounding a cell space</documentation>
          </annotation>
        </element>
      </sequence>
    </extension>
  </complexContent>
</complexType>
<complexType name="CellSpacePropertyType">
  <sequence minOccurs="0">
    <element ref="IndoorGML:CellSpace"/>
  </sequence>
```

```
  <attributeGroup ref="gml:AssociationAttributeGroup"/>
  <attributeGroup ref="gml:OwnershipAttributeGroup"/>
 </complexType>
 <element name="CellSpaceGeometry" substitutionGroup="gml:AbstractObject" type="IndoorGML:
CellSpaceGeometryType"/>
 <complexType name="CellSpaceGeometryType">
  <sequence>
   <element minOccurs="0" name="Geometry3D" type="gml:SolidPropertyType"/>
   <element minOccurs="0" name="Geometry2D" type="gml:SurfacePropertyType"/>
  </sequence>
 </complexType>
 <complexType name="CellSpaceGeometryPropertyType">
  <sequence>
   <element ref="IndoorGML:CellSpaceGeometry"/>
  </sequence>
 </complexType>
 <element name="DualSpaceLayer" substitutionGroup="gml:AbstractFeature" type="IndoorGML:
DualSpaceLayerType"/>
 <complexType name="DualSpaceLayerType">
  <complexContent>
   <extension base="gml:AbstractFeatureType">
    <sequence>
     <element name="isLogical" type="boolean"/>
     <element minOccurs="0" name="creationDate" type="dateTime"/>
     <element minOccurs="0" name="terminationDate" type="dateTime"/>
     <element name="isDirected" type="boolean"/>
     <element maxOccurs="unbounded" minOccurs="0" name="edgeMember">
      <annotation>
       <documentation>Edges of a dual space</documentation>
      </annotation>
      <complexType>
       <complexContent>
        <extension base="gml:AbstractFeatureMemberType">
         <sequence minOccurs="0">
          <element ref="IndoorGML:Edge"/>
         </sequence>
         <attributeGroup ref="gml:AssociationAttributeGroup"/>
        </extension>
       </complexContent>
      </complexType>
     </element>
     <element maxOccurs="unbounded" name="nodeMember">
      <annotation>
       <documentation>Nodes of a dual space</documentation>
      </annotation>
      <complexType>
       <complexContent>
        <extension base="gml:AbstractFeatureMemberType">
         <sequence minOccurs="0">
          <element ref="IndoorGML:Node"/>
         </sequence>
         <attributeGroup ref="gml:AssociationAttributeGroup"/>
        </extension>
       </complexContent>
      </complexType>
     </element>
    </sequence>
   </extension>
  </complexContent>
 </complexType>
 <complexType name="DualSpaceLayerPropertyType">
  <sequence minOccurs="0">
   <element ref="IndoorGML:DualSpaceLayer"/>
```

```xml
  </sequence>
  <attributeGroup ref="gml:AssociationAttributeGroup"/>
  <attributeGroup ref="gml:OwnershipAttributeGroup"/>
 </complexType>
 <element name="Edge" substitutionGroup="gml:AbstractFeature" type="IndoorGML:EdgeType"/>
 <complexType name="EdgeType">
  <complexContent>
   <extension base="gml:AbstractFeatureType">
    <sequence>
     <element minOccurs="0" name="geometry" type="gml:CurvePropertyType"/>
     <element name="weight" type="double"/>
     <element minOccurs="0" name="duality" type="IndoorGML:CellBoundaryPropertyType">
      <annotation>
       <documentation>Corresponding edge in the dual space</documentation>
       <appinfo>
        <reversePropertyName xmlns="http://www.opengis.net/gml/3.2">IndoorGML:
duality</reversePropertyName>
       </appinfo>
      </annotation>
     </element>
     <element maxOccurs="2" minOccurs="2" name="connects" type="IndoorGML:NodePropertyType">
      <annotation>
       <documentation>The two nodes connected by an edge</documentation>
       <appinfo>
        <reversePropertyName xmlns="http://www.opengis.net/gml/3.2">IndoorGML:
connects</reversePropertyName>
       </appinfo>
      </annotation>
     </element>
    </sequence>
   </extension>
  </complexContent>
 </complexType>
 <complexType name="EdgePropertyType">
  <sequence minOccurs="0">
   <element ref="IndoorGML:Edge"/>
  </sequence>
  <attributeGroup ref="gml:AssociationAttributeGroup"/>
  <attributeGroup ref="gml:OwnershipAttributeGroup"/>
 </complexType>
 <element name="ExternalObjectReference" substitutionGroup="gml:AbstractObject" type="IndoorGML:
ExternalObjectReferenceType"/>
 <complexType name="ExternalObjectReferenceType">
  <sequence>
   <element name="name" type="string"/>
   <element name="uri" type="anyURI"/>
  </sequence>
 </complexType>
 <complexType name="ExternalObjectReferencePropertyType">
  <sequence>
   <element ref="IndoorGML:ExternalObjectReference"/>
  </sequence>
 </complexType>
 <element name="ExternalReference" substitutionGroup="gml:AbstractObject" type="IndoorGML:
ExternalReferenceType"/>
 <complexType name="ExternalReferenceType">
  <sequence>
   <element name="externalObject" type="IndoorGML:ExternalObjectReferencePropertyType"/>
   <element name="informationSystem" type="gmd:URL_PropertyType"/>
  </sequence>
 </complexType>
 <complexType name="ExternalReferencePropertyType">
  <sequence>
```

```xml
    <element ref="IndoorGML:ExternalReference"/>
   </sequence>
  </complexType>
  <element name="IndoorFeatures" substitutionGroup="gml:AbstractFeature" type="IndoorGML:
IndoorFeaturesType"/>
  <complexType name="IndoorFeaturesType">
   <complexContent>
    <extension base="gml:AbstractFeatureType">
     <sequence>
      <element maxOccurs="unbounded" name="layers">
       <complexType>
        <complexContent>
         <extension base="gml:AbstractFeatureMemberType">
          <sequence minOccurs="0">
           <element ref="IndoorGML:ThematicLayer"/>
          </sequence>
          <attributeGroup ref="gml:AssociationAttributeGroup"/>
         </extension>
        </complexContent>
       </complexType>
      </element>
      <element maxOccurs="unbounded" minOccurs="0" name="layerConnections">
       <complexType>
        <complexContent>
         <extension base="gml:AbstractFeatureMemberType">
          <sequence minOccurs="0">
           <element ref="IndoorGML:InterLayerConnection"/>
          </sequence>
          <attributeGroup ref="gml:AssociationAttributeGroup"/>
         </extension>
        </complexContent>
       </complexType>
      </element>
     </sequence>
    </extension>
   </complexContent>
  </complexType>
  <complexType name="IndoorFeaturesPropertyType">
   <sequence minOccurs="0">
    <element ref="IndoorGML:IndoorFeatures"/>
   </sequence>
   <attributeGroup ref="gml:AssociationAttributeGroup"/>
   <attributeGroup ref="gml:OwnershipAttributeGroup"/>
  </complexType>
  <element name="InterLayerConnection" substitutionGroup="gml:AbstractFeature" type="IndoorGML:
InterLayerConnectionType">
   <annotation>
    <documentation>Describes the connection between two thematic layers.</documentation>
   </annotation>
  </element>
  <complexType name="InterLayerConnectionType">
   <complexContent>
    <extension base="gml:AbstractFeatureType">
     <sequence>
      <element name="typeOfTopoExpression" type="IndoorGML:TopoExpressionValueType">
       <annotation>
        <documentation>Describes the topological relationship between two layers (e.g. overlaps, contains, etc.).
</documentation>
       </annotation>
      </element>
      <element minOccurs="0" name="comment" type="string"/>
      <element maxOccurs="2" minOccurs="2" name="connectedLayers" type="IndoorGML:
ThematicLayerPropertyType"/>
```

```
      <element maxOccurs="2" minOccurs="0" name="connectedCells" type="IndoorGML:CellSpacePropertyType">
       <annotation>
        <documentation>Indicates the interconnected primal spaces</documentation>
       </annotation>
      </element>
      <element maxOccurs="2" minOccurs="0" name="connectedNodes" type="IndoorGML:NodePropertyType">
       <annotation>
        <documentation>Indicates the interconnected dual spaces</documentation>
       </annotation>
      </element>
     </sequence>
    </extension>
   </complexContent>
  </complexType>
  <complexType name="InterLayerConnectionPropertyType">
   <sequence minOccurs="0">
    <element ref="IndoorGML:InterLayerConnection"/>
   </sequence>
   <attributeGroup ref="gml:AssociationAttributeGroup"/>
   <attributeGroup ref="gml:OwnershipAttributeGroup"/>
  </complexType>
  <element name="Node" substitutionGroup="gml:AbstractFeature" type="IndoorGML:NodeType"/>
  <complexType name="NodeType">
   <complexContent>
    <extension base="gml:AbstractFeatureType">
     <sequence>
      <element minOccurs="0" name="geometry" type="gml:PointPropertyType"/>
      <element name="duality" type="IndoorGML:CellSpacePropertyType">
       <annotation>
        <documentation>Corresponding node in the dual space</documentation>
        <appinfo>
         <reversePropertyName xmlns="http://www.opengis.net/gml/3.2">IndoorGML:
duality</reversePropertyName>
        </appinfo>
       </annotation>
      </element>
      <element maxOccurs="unbounded" minOccurs="0" name="connects" type="IndoorGML:EdgePropertyType">
       <annotation>
        <documentation>Edges connected to the node</documentation>
        <appinfo>
         <reversePropertyName xmlns="http://www.opengis.net/gml/3.2">IndoorGML:
connects</reversePropertyName>
        </appinfo>
       </annotation>
      </element>
     </sequence>
    </extension>
   </complexContent>
  </complexType>
  <complexType name="NodePropertyType">
   <sequence minOccurs="0">
    <element ref="IndoorGML:Node"/>
   </sequence>
   <attributeGroup ref="gml:AssociationAttributeGroup"/>
   <attributeGroup ref="gml:OwnershipAttributeGroup"/>
  </complexType>
  <element name="PrimalSpaceLayer" substitutionGroup="gml:AbstractFeature" type="IndoorGML:
PrimalSpaceLayerType"/>
  <complexType name="PrimalSpaceLayerType">
   <complexContent>
    <extension base="gml:AbstractFeatureType">
     <sequence>
      <element minOccurs="0" name="creationDate" type="dateTime"/>
```

```
      <element minOccurs="0" name="terminationDate" type="dateTime"/>
      <element maxOccurs="unbounded" name="cellSpaceMember">
       <annotation>
        <documentation>Cell spaces of a primal space</documentation>
       </annotation>
       <complexType>
        <complexContent>
         <extension base="gml:AbstractFeatureMemberType">
          <sequence minOccurs="0">
           <element ref="IndoorGML:CellSpace"/>
          </sequence>
          <attributeGroup ref="gml:AssociationAttributeGroup"/>
         </extension>
        </complexContent>
       </complexType>
      </element>
      <element maxOccurs="unbounded" minOccurs="0" name="cellBoundaryMember">
       <annotation>
        <documentation>Cell boundaries of a primal space</documentation>
       </annotation>
       <complexType>
        <complexContent>
         <extension base="gml:AbstractFeatureMemberType">
          <sequence minOccurs="0">
           <element ref="IndoorGML:CellBoundary"/>
          </sequence>
          <attributeGroup ref="gml:AssociationAttributeGroup"/>
         </extension>
        </complexContent>
       </complexType>
      </element>
     </sequence>
    </extension>
   </complexContent>
  </complexType>
  <complexType name="PrimalSpaceLayerPropertyType">
   <sequence minOccurs="0">
    <element ref="IndoorGML:PrimalSpaceLayer"/>
   </sequence>
   <attributeGroup ref="gml:AssociationAttributeGroup"/>
   <attributeGroup ref="gml:OwnershipAttributeGroup"/>
  </complexType>
  <element name="ThematicLayer" substitutionGroup="gml:AbstractFeature" type="IndoorGML:
ThematicLayerType">
   <annotation>
    <documentation>A layer of specific theme aggregating a primal and/or a dual space of a given environment.
</documentation>
   </annotation>
  </element>
  <complexType name="ThematicLayerType">
   <complexContent>
    <extension base="gml:AbstractFeatureType">
     <sequence>
      <element name="semanticExtension" type="boolean">
       <annotation>
        <documentation>Indicates whether semantic information is associated (true) or not (false) to the primal space
of the thematic layer.</documentation>
       </annotation>
      </element>
      <element name="theme" type="IndoorGML:ThemeLayerValueType">
       <annotation>
        <documentation>Determines the theme of the layer (e.g topographic, logical, etc.).</documentation>
       </annotation>
```

```xml
     </element>
     <element minOccurs="0" name="primalSpace">
      <annotation>
       <documentation>Relates the primal spaces to the thematicLayer</documentation>
      </annotation>
      <complexType>
       <complexContent>
        <extension base="gml:AbstractFeatureMemberType">
         <sequence minOccurs="0">
          <element ref="IndoorGML:PrimalSpaceLayer"/>
         </sequence>
         <attributeGroup ref="gml:AssociationAttributeGroup"/>
        </extension>
       </complexContent>
      </complexType>
     </element>
     <element minOccurs="0" name="dualSpace">
      <annotation>
       <documentation>Relates the dual spaces to the thematicLayer.</documentation>
      </annotation>
      <complexType>
       <complexContent>
        <extension base="gml:AbstractFeatureMemberType">
         <sequence minOccurs="0">
          <element ref="IndoorGML:DualSpaceLayer"/>
         </sequence>
         <attributeGroup ref="gml:AssociationAttributeGroup"/>
        </extension>
       </complexContent>
      </complexType>
     </element>
    </sequence>
   </extension>
  </complexContent>
 </complexType>
 <complexType name="ThematicLayerPropertyType">
  <sequence minOccurs="0">
   <element ref="IndoorGML:ThematicLayer"/>
  </sequence>
  <attributeGroup ref="gml:AssociationAttributeGroup"/>
  <attributeGroup ref="gml:OwnershipAttributeGroup"/>
 </complexType>
 <simpleType name="ThemeLayerValueType">
  <union memberTypes="IndoorGML:ThemeLayerValueEnumerationType IndoorGML:ThemeLayerValueOtherType"/
>
 </simpleType>
 <simpleType name="ThemeLayerValueEnumerationType">
  <restriction base="string">
   <enumeration value="Tags"/>
   <enumeration value="Virtual"/>
   <enumeration value="Unknown"/>
   <enumeration value="Physical"/>
  </restriction>
 </simpleType>
 <simpleType name="ThemeLayerValueOtherType">
  <restriction base="string">
   <pattern value="other: \w{2,}"/>
  </restriction>
 </simpleType>
 <simpleType name="TopoExpressionValueType">
  <union memberTypes="IndoorGML:TopoExpressionValueEnumerationType IndoorGML:
TopoExpressionValueOtherType"/>
 </simpleType>
```

```
 <simpleType name="TopoExpressionValueEnumerationType">
  <restriction base="string">
   <enumeration value="Contains"/>
   <enumeration value="Overlaps"/>
   <enumeration value="Equals"/>
   <enumeration value="Within"/>
   <enumeration value="Crosses"/>
   <enumeration value="Other"/>
  </restriction>
 </simpleType>
 <simpleType name="TopoExpressionValueOtherType">
  <restriction base="string">
   <pattern value="other: \w{2,}"/>
  </restriction>
 </simpleType>
</schema>
```

Figure 3

10.2. IndoorGML Indoor Navigation Module

```
<?xml version="1.0" encoding="UTF-8"?><schema xmlns="http://www.w3.org/2001/XMLSchema" xmlns:IndoorGML=
"http://www.opengis.net/indoorgml/2.0/core" xmlns:gmd="http://www.isotc211.org/2005/gmd" xmlns:gml=
"http://www.opengis.net/gml/3.2" elementFormDefault="qualified" targetNamespace="http://www.opengis.net/
indoorgml/2.0/navi" version="2.0">
 <annotation>
  <appinfo>
   <gmlProfileSchema xmlns="http://www.opengis.net/gml/3.2">IndoorGML</gmlProfileSchema>
  </appinfo>
 </annotation>
 <import namespace="http://www.isotc211.org/2005/gmd" schemaLocation="http://schemas.opengis.net/iso/19139
/20070417/gmd/gmd.xsd"/>
 <import namespace="http://www.opengis.net/gml/3.2" schemaLocation="http://schemas.opengis.net/gml/3.2.1/
gml.xsd"/>
 <import namespace="http://www.opengis.net/indoorgml/2.0/core"/>
 <!--XML Schema document created by ShapeChange - http://shapechange.net/-->
 <element name="AbstractFreature" substitutionGroup="gml:AbstractFeature" type="IndoorGML:
AbstractFreatureType"/>
 <complexType name="AbstractFreatureType">
  <complexContent>
   <extension base="gml:AbstractFeatureType">
    <sequence/>
   </extension>
  </complexContent>
 </complexType>
 <complexType name="AbstractFreaturePropertyType">
  <sequence minOccurs="0">
   <element ref="IndoorGML:AbstractFreature"/>
  </sequence>
  <attributeGroup ref="gml:AssociationAttributeGroup"/>
  <attributeGroup ref="gml:OwnershipAttributeGroup"/>
 </complexType>
 <element name="CellBoundary" substitutionGroup="gml:AbstractFeature" type="IndoorGML:CellBoundaryType"/>
 <complexType name="CellBoundaryType">
  <complexContent>
   <extension base="gml:AbstractFeatureType">
    <sequence>
```

```
      <element minOccurs="0" name="cellBoundaryGeom" type="IndoorGML:CellBoundaryGeometryPropertyType"/
>
      <element minOccurs="0" name="externalReference" type="IndoorGML:ExternalReferencePropertyType"/>
      <element name="isVirtual" type="boolean"/>
      <element minOccurs="0" name="duality" type="IndoorGML:EdgePropertyType">
       <annotation>
        <appinfo>
         <reversePropertyName xmlns="http://www.opengis.net/gml/3.2">IndoorGML:
duality</reversePropertyName>
        </appinfo>
       </annotation>
      </element>
     </sequence>
    </extension>
   </complexContent>
  </complexType>
  <complexType name="CellBoundaryPropertyType">
   <sequence minOccurs="0">
    <element ref="IndoorGML:CellBoundary"/>
   </sequence>
   <attributeGroup ref="gml:AssociationAttributeGroup"/>
   <attributeGroup ref="gml:OwnershipAttributeGroup"/>
  </complexType>
  <element name="CellBoundaryGeometry" substitutionGroup="gml:AbstractObject" type="IndoorGML:
CellBoundaryGeometryType"/>
  <complexType name="CellBoundaryGeometryType">
   <sequence>
    <element minOccurs="0" name="Geometry2D" type="gml:SurfacePropertyType"/>
    <element minOccurs="0" name="Geometry1D" type="gml:CurvePropertyType"/>
   </sequence>
  </complexType>
  <complexType name="CellBoundaryGeometryPropertyType">
   <sequence>
    <element ref="IndoorGML:CellBoundaryGeometry"/>
   </sequence>
  </complexType>
  <element name="CellSpace" substitutionGroup="IndoorGML:AbstractFeature" type="IndoorGML:CellSpaceType"/>
  <complexType name="CellSpaceType">
   <complexContent>
    <extension base="IndoorGML:AbstractFeatureType">
     <sequence>
      <element minOccurs="0" name="cellSpaceGeom" type="IndoorGML:CellSpaceGeometryPropertyType"/>
      <element minOccurs="0" name="externalReference" type="IndoorGML:ExternalReferencePropertyType"/>
      <element minOccurs="0" name="level" type="string"/>
      <element name="PoI" type="boolean"/>
      <element minOccurs="0" name="name" type="string"/>
      <element minOccurs="0" name="duality" type="IndoorGML:NodePropertyType">
       <annotation>
        <appinfo>
         <reversePropertyName xmlns="http://www.opengis.net/gml/3.2">IndoorGML:
duality</reversePropertyName>
        </appinfo>
       </annotation>
      </element>
      <element maxOccurs="unbounded" minOccurs="0" name="boundedBy" type="IndoorGML:
CellBoundaryPropertyType">
       <annotation>
        <documentation>Cell boundaries that are bounding a cell space</documentation>
       </annotation>
      </element>
     </sequence>
    </extension>
   </complexContent>
```

```xml
  </complexType>
  <complexType name="CellSpacePropertyType">
    <sequence minOccurs="0">
      <element ref="IndoorGML:CellSpace"/>
    </sequence>
    <attributeGroup ref="gml:AssociationAttributeGroup"/>
    <attributeGroup ref="gml:OwnershipAttributeGroup"/>
  </complexType>
  <element name="CellSpaceGeometry" substitutionGroup="gml:AbstractObject" type="IndoorGML:
CellSpaceGeometryType"/>
  <complexType name="CellSpaceGeometryType">
    <sequence>
      <element minOccurs="0" name="Geometry3D" type="gml:SolidPropertyType"/>
      <element minOccurs="0" name="Geometry2D" type="gml:SurfacePropertyType"/>
    </sequence>
  </complexType>
  <complexType name="CellSpaceGeometryPropertyType">
    <sequence>
      <element ref="IndoorGML:CellSpaceGeometry"/>
    </sequence>
  </complexType>
  <element name="Edge" substitutionGroup="gml:AbstractFeature" type="IndoorGML:EdgeType"/>
  <complexType name="EdgeType">
    <complexContent>
      <extension base="gml:AbstractFeatureType">
        <sequence>
          <element minOccurs="0" name="geometry" type="gml:CurvePropertyType"/>
          <element name="weight" type="double"/>
          <element maxOccurs="2" minOccurs="2" name="connects" type="IndoorGML:NodePropertyType">
            <annotation>
              <documentation>The two nodes connected by an edge</documentation>
              <appinfo>
                <reversePropertyName xmlns="http://www.opengis.net/gml/3.2">IndoorGML:
connects</reversePropertyName>
              </appinfo>
            </annotation>
          </element>
          <element minOccurs="0" name="duality" type="IndoorGML:CellBoundaryPropertyType">
            <annotation>
              <appinfo>
                <reversePropertyName xmlns="http://www.opengis.net/gml/3.2">IndoorGML:
duality</reversePropertyName>
              </appinfo>
            </annotation>
          </element>
        </sequence>
      </extension>
    </complexContent>
  </complexType>
  <complexType name="EdgePropertyType">
    <sequence minOccurs="0">
      <element ref="IndoorGML:Edge"/>
    </sequence>
    <attributeGroup ref="gml:AssociationAttributeGroup"/>
    <attributeGroup ref="gml:OwnershipAttributeGroup"/>
  </complexType>
  <element name="ExternalObjectReference" substitutionGroup="gml:AbstractObject" type="IndoorGML:
ExternalObjectReferenceType"/>
  <complexType name="ExternalObjectReferenceType">
    <sequence>
      <element name="uri" type="anyURI"/>
      <element name="name" type="string"/>
    </sequence>
```

```
 </complexType>
 <complexType name="ExternalObjectReferencePropertyType">
  <sequence>
   <element ref="IndoorGML:ExternalObjectReference"/>
  </sequence>
 </complexType>
 <element name="ExternalReference" substitutionGroup="gml:AbstractObject" type="IndoorGML:
ExternalReferenceType"/>
 <complexType name="ExternalReferenceType">
  <sequence>
   <element name="externalObject">
    <complexType>
     <complexContent>
      <extension base="gml:AbstractMemberType">
       <sequence minOccurs="0">
        <element ref="IndoorGML:ExternalObjectReferenceType"/>
       </sequence>
       <attributeGroup ref="gml:AssociationAttributeGroup"/>
      </extension>
     </complexContent>
    </complexType>
   </element>
   <element name="informationSystem" type="gmd:URL_PropertyType"/>
  </sequence>
 </complexType>
 <complexType name="ExternalReferencePropertyType">
  <sequence>
   <element ref="IndoorGML:ExternalReference"/>
  </sequence>
 </complexType>
 <element name="GeneralSpace" substitutionGroup="IndoorGML:NavigableSpace" type="IndoorGML:
GeneralSpaceType"/>
 <complexType name="GeneralSpaceType">
  <complexContent>
   <extension base="IndoorGML:NavigableSpaceType">
    <sequence>
     <element name="function" type="IndoorGML:GeneralSpaceFunctionTypeType"/>
    </sequence>
   </extension>
  </complexContent>
 </complexType>
 <complexType name="GeneralSpacePropertyType">
  <sequence minOccurs="0">
   <element ref="IndoorGML:GeneralSpace"/>
  </sequence>
  <attributeGroup ref="gml:AssociationAttributeGroup"/>
  <attributeGroup ref="gml:OwnershipAttributeGroup"/>
 </complexType>
 <simpleType name="GeneralSpaceFunctionType">
  <union memberTypes="IndoorGML:GeneralSpaceFunctionEnumerationType IndoorGML:
GeneralSpaceFunctionOtherType"/>
 </simpleType>
 <simpleType name="GeneralSpaceFunctionEnumerationType">
  <restriction base="string">
   <enumeration value="Administration"/>
   <enumeration value="Business, trade"/>
   <enumeration value="Education, training"/>
   <enumeration value="Recreation"/>
   <enumeration value="Laboratory"/>
   <enumeration value="Storage"/>
   <enumeration value="Security"/>
  </restriction>
 </simpleType>
```

```xml
 <simpleType name="GeneralSpaceFunctionOtherType">
  <restriction base="string">
   <pattern value="other: \w{2,}"/>
  </restriction>
 </simpleType>
 <simpleType name="LocomotionAccessType">
  <union memberTypes="IndoorGML:LocomotionAccessEnumerationType IndoorGML:
LocomotionAccessOtherType"/>
 </simpleType>
 <simpleType name="LocomotionAccessEnumerationType">
  <restriction base="string">
   <enumeration value="Walking"/>
   <enumeration value="Flying"/>
   <enumeration value="Rolling"/>
   <enumeration value="Unspecified"/>
  </restriction>
 </simpleType>
 <simpleType name="LocomotionAccessOtherType">
  <restriction base="string">
   <pattern value="other: \w{2,}"/>
  </restriction>
 </simpleType>
 <element name="NavigableBoundary" substitutionGroup="IndoorGML:CellBoundary" type="IndoorGML:
NavigableBoundaryType"/>
 <complexType name="NavigableBoundaryType">
  <complexContent>
   <extension base="IndoorGML:CellBoundaryType">
    <sequence>
     <element name="NavigableBoundaryFunction" type="IndoorGML:NavigableBoundaryFunctionTypeType"/>
     <element minOccurs="0" name="boundaryOrientation" type="boolean"/>
    </sequence>
   </extension>
  </complexContent>
 </complexType>
 <complexType name="NavigableBoundaryPropertyType">
  <sequence minOccurs="0">
   <element ref="IndoorGML:NavigableBoundary"/>
  </sequence>
  <attributeGroup ref="gml:AssociationAttributeGroup"/>
  <attributeGroup ref="gml:OwnershipAttributeGroup"/>
 </complexType>
 <simpleType name="NavigableBoundaryFunctionType">
  <union memberTypes="IndoorGML:NavigableBoundaryFunctionEnumerationType IndoorGML:NavigableBoundary
FunctionOtherType"/>
 </simpleType>
 <simpleType name="NavigableBoundaryFunctionEnumerationType">
  <restriction base="string">
   <enumeration value="AncorBoundary"/>
   <enumeration value="ConnectionBoundary"/>
  </restriction>
 </simpleType>
 <simpleType name="NavigableBoundaryFunctionOtherType">
  <restriction base="string">
   <pattern value="other: \w{2,}"/>
  </restriction>
 </simpleType>
 <element name="NavigableSpace" substitutionGroup="IndoorGML:CellSpace" type="IndoorGML:
NavigableSpaceType"/>
 <complexType name="NavigableSpaceType">
  <complexContent>
   <extension base="IndoorGML:CellSpaceType">
    <sequence>
     <element name="locomotionType" type="IndoorGML:LocomotionAccessType"/>
```

```xml
      </sequence>
    </extension>
  </complexContent>
 </complexType>
 <complexType name="NavigableSpacePropertyType">
  <sequence minOccurs="0">
    <element ref="IndoorGML:NavigableSpace"/>
  </sequence>
  <attributeGroup ref="gml:AssociationAttributeGroup"/>
  <attributeGroup ref="gml:OwnershipAttributeGroup"/>
 </complexType>
 <element name="Node" substitutionGroup="gml:AbstractFeature" type="IndoorGML:NodeType"/>
 <complexType name="NodeType">
  <complexContent>
    <extension base="gml:AbstractFeatureType">
      <sequence>
        <element minOccurs="0" name="geometry" type="gml:PointPropertyType"/>
        <element maxOccurs="unbounded" minOccurs="0" name="connects" type="IndoorGML:EdgePropertyType">
          <annotation>
            <documentation>Edges connected to the node</documentation>
            <appinfo>
              <reversePropertyName xmlns="http://www.opengis.net/gml/3.2">IndoorGML:
connects</reversePropertyName>
            </appinfo>
          </annotation>
        </element>
        <element name="duality" type="IndoorGML:CellSpacePropertyType">
          <annotation>
            <appinfo>
              <reversePropertyName xmlns="http://www.opengis.net/gml/3.2">IndoorGML:
duality</reversePropertyName>
            </appinfo>
          </annotation>
        </element>
      </sequence>
    </extension>
  </complexContent>
 </complexType>
 <complexType name="NodePropertyType">
  <sequence minOccurs="0">
    <element ref="IndoorGML:Node"/>
  </sequence>
  <attributeGroup ref="gml:AssociationAttributeGroup"/>
  <attributeGroup ref="gml:OwnershipAttributeGroup"/>
 </complexType>
 <element name="NonNavigableBoundary" substitutionGroup="IndoorGML:CellBoundary" type="IndoorGML:
NonNavigableBoundaryType"/>
 <complexType name="NonNavigableBoundaryType">
  <complexContent>
    <extension base="IndoorGML:CellBoundaryType">
      <sequence/>
    </extension>
  </complexContent>
 </complexType>
 <complexType name="NonNavigableBoundaryPropertyType">
  <sequence minOccurs="0">
    <element ref="IndoorGML:NonNavigableBoundary"/>
  </sequence>
  <attributeGroup ref="gml:AssociationAttributeGroup"/>
  <attributeGroup ref="gml:OwnershipAttributeGroup"/>
 </complexType>
 <element name="NonNavigableSpace" substitutionGroup="IndoorGML:CellSpace" type="IndoorGML:
NonNavigableSpaceType"/>
```

```
 <complexType name="NonNavigableSpaceType">
  <complexContent>
   <extension base="IndoorGML:CellSpaceType">
    <sequence/>
   </extension>
  </complexContent>
 </complexType>
 <complexType name="NonNavigableSpacePropertyType">
  <sequence minOccurs="0">
   <element ref="IndoorGML:NonNavigableSpace"/>
  </sequence>
  <attributeGroup ref="gml:AssociationAttributeGroup"/>
  <attributeGroup ref="gml:OwnershipAttributeGroup"/>
 </complexType>
 <element name="ObjectSpace" substitutionGroup="IndoorGML:NonNavigableSpace" type="IndoorGML:
ObjectSpaceType"/>
 <complexType name="ObjectSpaceType">
  <complexContent>
   <extension base="IndoorGML:NonNavigableSpaceType">
    <sequence>
     <element minOccurs="0" name="description" type="string"/>
     <element minOccurs="0" name="containedFeatures" type="integer"/>
    </sequence>
   </extension>
  </complexContent>
 </complexType>
 <complexType name="ObjectSpacePropertyType">
  <sequence minOccurs="0">
   <element ref="IndoorGML:ObjectSpace"/>
  </sequence>
  <attributeGroup ref="gml:AssociationAttributeGroup"/>
  <attributeGroup ref="gml:OwnershipAttributeGroup"/>
 </complexType>
 <element name="Route" substitutionGroup="gml:AbstractFeature" type="IndoorGML:RouteType"/>
 <complexType name="RouteType">
  <complexContent>
   <extension base="gml:AbstractFeatureType">
    <sequence>
     <element minOccurs="0" name="creationDate" type="dateTime"/>
     <element maxOccurs="unbounded" minOccurs="2" name="routeNode" type="IndoorGML:NodePropertyType"/
>
     <element maxOccurs="unbounded" name="routeEdge" type="IndoorGML:EdgePropertyType"/>
    </sequence>
   </extension>
  </complexContent>
 </complexType>
 <complexType name="RoutePropertyType">
  <sequence minOccurs="0">
   <element ref="IndoorGML:Route"/>
  </sequence>
  <attributeGroup ref="gml:AssociationAttributeGroup"/>
  <attributeGroup ref="gml:OwnershipAttributeGroup"/>
 </complexType>
 <element name="TransferSpace" substitutionGroup="IndoorGML:NavigableSpace" type="IndoorGML:
TransferSpaceType"/>
 <complexType name="TransferSpaceType">
  <complexContent>
   <extension base="IndoorGML:NavigableSpaceType">
    <sequence>
     <element name="function" type="IndoorGML:TransferSpaceFunctionTypeType"/>
     <element name="type" type="IndoorGML:TransferSpaceTypeType"/>
    </sequence>
   </extension>
```

```
    </complexContent>
  </complexType>
  <complexType name="TransferSpacePropertyType">
    <sequence minOccurs="0">
      <element ref="IndoorGML:TransferSpace"/>
    </sequence>
    <attributeGroup ref="gml:AssociationAttributeGroup"/>
    <attributeGroup ref="gml:OwnershipAttributeGroup"/>
  </complexType>
  <simpleType name="TransferSpaceFunctionType">
    <union memberTypes="IndoorGML:TransferSpaceFunctionEnumerationType IndoorGML:TransferSpaceFunctionO
therType"/>
  </simpleType>
  <simpleType name="TransferSpaceFunctionEnumerationType">
    <restriction base="string">
      <enumeration value="ConnectionSpace"/>
      <enumeration value="AnchorSpace"/>
    </restriction>
  </simpleType>
  <simpleType name="TransferSpaceFunctionOtherType">
    <restriction base="string">
      <pattern value="other: \w{2,}"/>
    </restriction>
  </simpleType>
  <simpleType name="TransferSpaceTypeType">
    <union memberTypes="IndoorGML:TransferSpaceTypeEnumerationType IndoorGML:
TransferSpaceTypeOtherType"/>
  </simpleType>
  <simpleType name="TransferSpaceTypeEnumerationType">
    <restriction base="string">
      <enumeration value="Door"/>
      <enumeration value="Window"/>
    </restriction>
  </simpleType>
  <simpleType name="TransferSpaceTypeOtherType">
    <restriction base="string">
      <pattern value="other: \w{2,}"/>
    </restriction>
  </simpleType>
</schema>
```

Figure 4

11

EXAMPLE OF CODELIST

11 EXAMPLE OF CODELIST

For mapping indoor space features to IndoorNavigation module classes, space features have to be classified by functions and usages of space. Each space class has attributes for functions, usages, and classes of space. Because misspellings or different names for the same notion brings the problems in data interoperability, in order to overcome the typo errors, an example of CodeList is provided in this annex to specify the attribute values including space class types, space function types and space usage types. The CodeLists are defined from OmniClass (Table 13 and Table 14) created and used by the North American architectural, engineering and construction (AEC) industry.

11.1. GeneralSpace

11.1.1. GeneralSpaceClassType

Code list derived from OmniClass Table 13 and CityGML: * 1000 — Administration * 1010 — Business, trade * 1020 — Education, training * 1030 — Recreation * 1040 — Art, performance * 1050 — Healthcare * 1060 — Laboratory * 1070 — Service * 1080 — Production * 1090 — Storage * 1100 — Security * 1110 — Accommodation, Waste management

11.1.2. GeneralSpaceFunctionType

Code list derived from OmniClass Table 13: * 1000 — Elevator Machine Room * 1010 — Fire Command Center * 1020 — Men’s Restroom * 1030 — Unisex Restroom * 1040 — Refrigerant Machinery Room * 1050 — Incinerator Room * 1060 — Gas Room * 1070 — Liquid Use, Dispensing and Mixing Room * 1080 — Electrical Room * 1090 — Telecommunications Room * 1100 — Hazardous Waste Storage * 1110 — Building Manager Office * 1120 — Guard Stations * 1130 — Women’s Restroom * 1140 — Furnace Room * 1150 — Fuel Room * 1160 — Liquid Storage Room * 1170 — Hydrogen Cutoff Room * 1180 — Switch Room * 1190 — Classrooms * 1200 — Assembly Hall * 1210 — Physics Teaching Laboratory * 1220 — Open Class Laboratory * 1230 — Laboratory Service Space * 1240 — Computer Lab * 1250 — Training Support Space * 1260 — Study Room * 1270 — Evidence Room * 1280 — Witness Stand * 1290 — Robing Area * 1300 — Council Chambers * 1310 — Armory * 1320 — General Performance Spaces * 1330 — Orchestra Pit * 1340 — Performance Hall * 1350 — Audience Space * 1360 — Audience Seating Space * 1370 — Projection Booth * 1380 — Stage Wings * 1390 — Art Gallery * 1400 — Sculpture Garden * 1410 — Recording Studio * 1420 — Photo Lab * 1430 — Library * 1440 — Mediation Chapel * 1450 — Reflection Space * 1460 — Chapel * 1470 — Shrine * 1480 — Confessional Space * 1490 — Tabernacle * 1500 — Choir Loft * 1510 — Marriage Sanctuary * 1520 — Mental Health Quiet Room * 1530 — Bone Densitometry Room * 1540 — CT Simulator Room * 1550 — Head Radiographic Room * 1560 — Mobile Imaging System Alcove * 1570 — MRI System Component Room * 1580 — PET/CT Simulator

Room * 1590 — Radiographic Room * 1600 — Stereotactic Mammography Room * 1610 — Ultrasound/Optical Coherence Tomography Room * 1620 — Angiographic Control Room * 1630 — Angiographic Procedure Control Area * 1640 — Silver Collection Area * 1650 — Computer Image Processing Area * 1660 — CT Control Area * 1670 — Image Quality Control Room * 1680 — X-Ray, Plane Film Storage Space * 1690 — MRI Control Room * 1700 — MRI Viewing Room * 1710 — Radiographic Control Room * 1720 — Tele-Radiology/Tele-Medicine Room * 1730 — Radiation Diagnostic and Therapy Spaces * 1740 — Health Physics Laboratory * 1750 — Linear Accelerator Entrance Maze, Healthcare * 1760 — Radioactive Waste Storage Room, Healthcare * 1770 — Nuclear Medicine Scanning Room * 1780 — Patient Dose/Thyroid Uptake Room * 1790 — Radiopharmacy * 1800 — Radiation Therapy, Mold Fabrication Shop * 1810 — Hearth and Lung Diagnostic and Treatment Spaces * 1820 — Cardiac Catheter Instrument Room * 1830 — Cardiac Catheter Control Room * 1840 — Cardiac Electrophysiology Room * 1850 — Echocardiograph Room * 1860 — Extended Pulmonary Function Testing Laboratory * 1870 — Pacemaker ICD Interrogation Room * 1880 — Procedure Viewing Area * 1890 — Pulmonary Function Treadmill Room * 1900 — Respiratory Therapy Clean-up Room * 1910 — General Diagnostic Procedure and Treatment Spaces * 1920 — Endoscopy/Gastroenterology Spaces * 1930 — Clinical Laboratory Spaces * 1940 — Pharmacy Spaces * 1950 — Rehabilitation Spaces * 1960 — Medical Research and Development Spaces * 1970 — Chemistry Laboratories * 1980 — Physical Sciences Laboratories * 1990 — Earth and Environmental Sciences Laboratories * 2000 — Psychology Laboratories * 2010 — Dry Laboratories * 2020 — Wet Laboratories * 2030 — Biosciences Laboratories * 2040 — Astronomy Laboratories * 2050 — Forensics Laboratories * 2060 — Bench Laboratories * … **(Refer to the full standard implementation schema for other classifications)**

11.1.3. GeneralSpaceUsageType

Code list identically specified as GeneralSpaceFunctionType.

11.2. TransitionSpace

11.2.1. TransitionSpaceClassType

- 1000 — Horizontal Transition
- 1010 — Vertical Transition

11.2.2. TransitionSpaceFunctionType

- 1000 — Corridor
- 1010 — Breezeway

- 1020 — Box Lobby
- 1030 — Elevator Lobby
- 1040 — Landing
- 1050 — Aisle
- 1060 — Ramp
- 1070 — Concourse
- 1080 — Moving walkway
- 1090 — Entry Lobby
- 1100 — Jet way
- 1110 — Elevator Shaft
- 1120 — Stair
- 1130 — Chute

11.2.3. TransitionSpaceUsageType

Code list identically specified as TransitionSpaceFunctionType.

11.3. ConnectionSpace

11.3.1. ConnectionSpaceClassType

- 1000 — Door
- 1010 — Vestibule
- 1020 — Sally port

11.3.2. ConnectionSpaceFunctionType

- 1000 — Door
- 1010 — Vestibule
- 1060 — Sally port

11.3.3. ConnectionSpaceUsageType

Code list identically specified as ConnectionSpaceFunctionType.

11.4. AnchorSpace

11.4.1. AnchorSpaceClassType

- 1000 — Vestibule
- 1020 — Gate

11.4.2. AnchorSpaceFunctionType

- 1000 — Entry Vestibule
- 1010 — Exterior door
- 1020 — Gate
- 1030 — Emergency door

11.4.3. AnchorSpaceUsageType

Code list identically specified as AnchorSpaceFunctionType.

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USE-CASES OF INDOORGML 2 — PART2A

12 USE-CASES OF INDOORGML 2 — PART2A

12.1. Use-Case 1: Conversion from IndoorGML 1.1 to IndoorGML 2

- Contributor: Ki-Joune Li

12.2. Use-Case 2: Conversion from IFC to IndoorGML 2

- Contributor: Abdoulaye Diakité

12.3. Use-Case 3: Mapping of ISO 19164 in IndoorGML 2 – Part 2a

- Contributors: Ki-Joune Li & Abdoulaye Diakité

12.4. Use-Case 4: Implementation of omlox in IndoorGML 2 – Part 2a

12.5. Use-Case 5: Indoor Navigation Service for Visually Impaired Person

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REVISION HISTORY

13 REVISION HISTORY

Table 3

Date	Release	Author	Paragraph modified	Description
2025-08-06	2.0.0	Sisi Zlatanova, Abdoulaye Diakité, Taehoon Kim, Ki-Joune Li	All	First draft of IndoorGML 2.0 Part 2a (XML Encoding)

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